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Project: Is Weekend Bitcoin Movement Fully Priced In at Monday's Open?

1. Hypothesis Pitch

I have followed cryptocurrency markets for a long time, and I noticed that Bitcoin trades 24 hours a day, seven days a week, while US stocks trade in a regular session from 09:30 to 16:00 New York time, Monday to Friday. Several listed companies are now closely tied to Bitcoin, either as miners (MARA, RIOT) or as holders of Bitcoin on their balance sheet (MSTR). Every weekend, Bitcoin keeps trading for about 65 hours while these stocks are outside their regular session.

The weekend Bitcoin move should show up in Monday's opening prices, but a trader who buys at the open cannot earn that jump. If a stock closes at $100 on Friday, opens at $106 and closes at $107, buying at the open earns about 0.9%, because the first 6% happened before the market opened. My hypothesis is about what happens after the open. I expect a positive Bitcoin return between Friday 16:00 and Monday 09:00 New York time to be followed by a positive open-to-close return on Monday for an equal-weight basket of MSTR, MARA and RIOT. That would mean the opening price takes in only part of the weekend information. The signal ends at 09:00 so that it is known before the market opens.

There are two reasons this could happen. Fewer investors follow markets at the weekend, and on Monday morning they have other news to deal with, so the open may not fully reflect what Bitcoin did. It is also not obvious how much a Bitcoin move should change these companies' values, since MSTR's value depends on its Bitcoin holdings and a miner's profit depends on the Bitcoin price minus its costs. Investors may need part of the day to work this out. Pre-market trading works against the idea, because some adjustment can happen before 09:30, but volume there is usually thin.

I would reject the hypothesis if Bitcoin explains the opening gap but not the Monday drift, if buying only after positive Bitcoin weekends does no better than buying every Monday once trading costs are included, if an effect found in 2021–2023 does not hold in the 2024–2025 test period, or if a handful of extreme weekends drive the result.

2. Data Pipeline

2.1. Sources and universe

Data	Source	Coverage
Bitcoin	Binance spot BTC/USDT public archive, 30-minute candles	Dec 2020–Dec 2025, 89,094 candles
Equities	yfinance daily OHLC with splits and dividends (`auto_adjust=False`) for MSTR, MARA, RIOT	1,277 sessions each
Calendar	`pandas_market_calendars` NYSE schedule	1,277 sessions, 11 early closes

Table 1: Data sources.

Binance was used because yfinance does not offer several years of 30-minute Bitcoin data, while Binance's archive is free and publishes a checksum for every file. BTC/USDT is a proxy for BTC/USD, so it carries the risk of USDT moving away from $1 and of problems specific to one exchange. Daily stock data is enough because the strategy only trades at the open and the close.

The universe is fixed at three companies whose value was tied to Bitcoin for the whole sample.

MSTR bought its first 21,454 BTC for $250 million on 11 August 2020, and MARA and RIOT were both running Bitcoin mining operations by 2018, according to their filings (see References).

2.2. Timing

Timestamps are stored in UTC and converted with the `America/New_York` time zone, so daylight-saving changes are handled without fixed offsets. A 30-minute candle is only known when it ends, so each candle is indexed by its end time and the price at time T is the close of the candle that ended exactly at T. The signal runs from 16:00 on the prior trading day to 09:00 on the entry day. If either endpoint candle is missing, the event is dropped.

2.3. Eligible events

Every Monday in 2021–2025 is a candidate weekend event, and every Tuesday to Friday is a candidate weeknight event for the comparison. An event is kept only if the entry day and the prior calendar weekday are both full NYSE sessions and both Bitcoin endpoints exist. All exclusions are logged (Table 2).

	Weekend (Mondays)		Weeknight (Tue–Fri)	
	2021–23	2024–25	2021–23	2024–25
Candidate days	156	105	625	418
Market closed (entry or prior day)	24	12	32	30
Short session (entry or prior day)	4	2	0	4
Bitcoin endpoint candle missing	0	0	1	0
Eligible events	**128**	**91**	**592**	**384**

Table 2: Candidate days, exclusions and eligible events.

2.4. Cleaning

- **Timestamp units.** Binance changed from millisecond to microsecond timestamps in January 2025. The unit is set per file, and the run stops if any timestamp falls outside the month in the file name.
- **Incomplete candles.** Five candles end early because of exchange outages, so their close is not the price at the boundary. They are flagged and never used as endpoints. One was the 09:00 cutoff candle on 24 March 2023, so that event was dropped.
- **Missing data.** 42 Bitcoin candles are missing across 9 outages, and each stock has exactly one row per NYSE session. Nothing is forward-filled, and a missing price after entry would stop the backtest rather than count as a zero return.
- **Corporate actions.** yfinance Open and Close with `auto_adjust=False` are already split-adjusted. None of the three companies paid a cash dividend, and MSTR's 10-for-1 split on 8 August 2024 shows a normal +4.4% opening gap instead of a false −90% jump.

3. Strategy and Backtest

3.1. Strategy

On each eligible Monday, if Bitcoin's return from Friday 16:00 to Monday 09:00 New York time is positive, the strategy buys an equal-weight basket of MSTR, MARA and RIOT at the regular open and sells it at the same day's close. Otherwise it holds cash, which earns zero. There is no leverage,

no short selling and no overnight position. Trading costs are set at 10 basis points (0.1%) on each purchase and each sale, so the strategy pays 0.1% above the opening price and receives 0.1% below the closing price. A day on which the price does not move therefore loses about 0.2%, and the net return of a trade is $(\text{Close}/\text{Open})(1-c)/(1+c)-1$ with $c = 0.001$. The benchmark buys the same basket on every eligible Monday under the same rules. The design was fixed on 2021–2023 data and frozen before the 2024–2025 test period was evaluated. All results are calculated from a return for every trading day in the period. Trade days get the trade's net return and cash days get 0%, so the idle days are counted rather than ignored. The Sharpe ratio measures return per unit of risk. It is the average daily return divided by the standard deviation of daily returns, multiplied by $\sqrt{252}$ to express it per year (about 252 trading days), and it assumes a risk-free rate of zero.

3.2. Key results

Period	Strategy	Trades	Exposure	Gross cum.	Net cum.	Sharpe	Max DD
In-sample 2021–23	BTC signal	60	8.0%	−8.5%	−18.8%	−0.16	−31.9%
	Every Monday	128	17.0%	+3.7%	−19.7%	−0.05	−50.8%
Out-of-sample 2024–25	BTC signal	53	10.6%	+64.0%	+47.5%	0.82	−18.9%
	Every Monday	91	18.1%	+72.1%	+43.5%	0.67	−31.3%

Table 3: Backtest results at 10 bps per side. Max DD is maximum drawdown.

Before costs the benchmark earned more than the signal out-of-sample (72.1% against 64.0%). The signal finishes ahead only after costs, because it paid for 53 trades instead of 91.

3.3. Gap versus drift

Bitcoin's weekend return explains the opening gap almost one-for-one, with a slope of 0.99 in-sample and 1.04 out-of-sample and $R^2 \approx 0.87$ in both (Figure 1). It does not explain the Monday drift. The slope is 0.03 in-sample ($p = 0.87$) and −0.30 out-of-sample ($p = 0.41$), and the negative value is mostly due to one weekend, 5 August 2024, when Bitcoin fell about 20% and the basket then rose about 19% during Monday. The average drift after positive Bitcoin weekends minus the average after the others is −0.31% in-sample (95% bootstrap interval $[-2.2\%, +1.6\%]$) and +0.82% out-of-sample ($[-1.6\%, +3.0\%]$), and both intervals include zero. Weeknights show the same pattern.

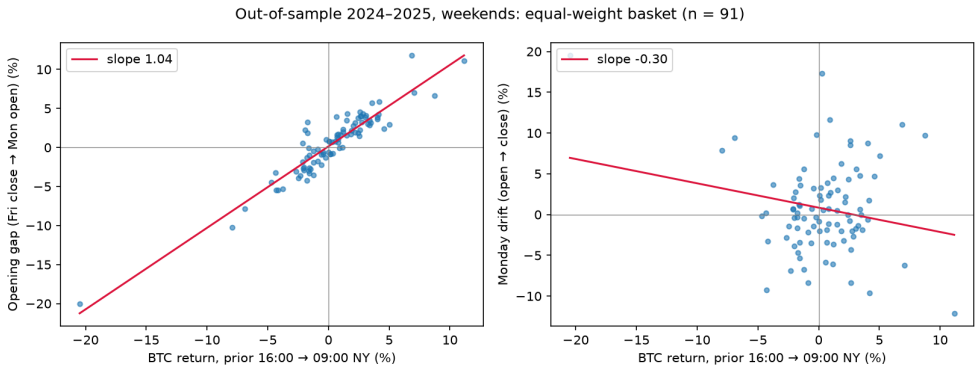


Figure 1: Out-of-sample weekends (2024–2025). Left: Bitcoin's Friday 16:00 to Monday 09:00 return against the basket's opening gap. Right: against the Monday open-to-close drift.

3.4. Is the timing better than chance?

The average net return of the strategy minus the benchmark per event is −0.03% out-of-sample (95% interval [−0.8%, +0.65%]). As a further check, the same number of Mondays that the signal traded was chosen at random 10,000 times under the same costs (Figure 2). The signal beats 37% of random selections in-sample and 76% out-of-sample, which is well inside the range produced by chance. Using Bitcoin to choose Mondays did no better than choosing them at random, so the signal's result came from the number of days it traded.

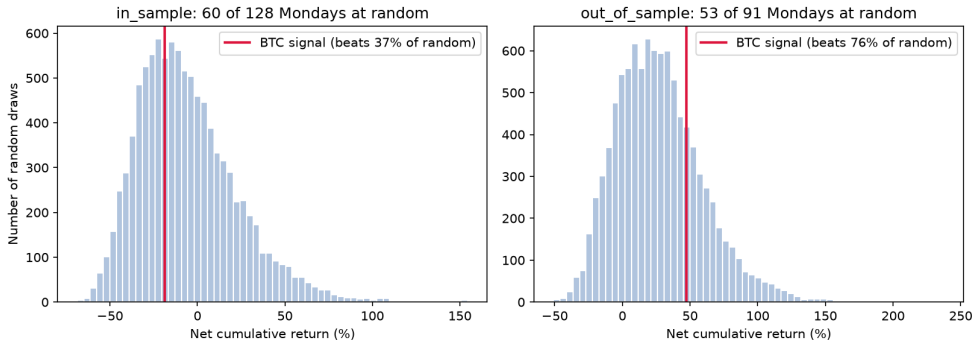


Figure 2: Net cumulative return of 10,000 random selections of eligible Mondays (same number of trades as the signal) and the Bitcoin signal (red line).

3.5. Robustness

The result is not stable. Out-of-sample, the signal returned +58.3% in 2024 and −6.9% in 2025, and removing its five best days turns +47.5% into −14.3%. A hindsight-only regression that adds Bitcoin's own Monday 09:30–16:00 return explains much of the drift (coefficient 1.74, $R^2 = 0.64$ out-of-sample), while the weekend return stays insignificant. The stocks move with Bitcoin during Monday rather than catching up on the weekend. The evidence therefore does not support the hypothesis. The opening price absorbs the weekend move and leaves no drift to trade.

4. Methodology Defense

Each main design choice is listed with the reason for it and what would go wrong if its assumption failed. All of them were fixed before the out-of-sample period was evaluated.

Choice	Why	If the assumption is wrong
Universe: MSTR, MARA, RIOT, equal weights	All three depended directly on Bitcoin for the whole sample and are liquid at the open. Equal weights leave nothing to optimise.	If a company's link to Bitcoin weakened, unrelated moves would enter the test. The result does not extend to other crypto stocks.
Split: 2021–23 in-sample, 2024–25 out-of-sample	A chronological split keeps the test period in the future. The rule has no fitted parameter, so no separate validation year was needed.	Spot Bitcoin ETFs launched in January 2024, so the test covers a different market. It is a fair test, but of one regime only.
Signal: Bitcoin return from Friday 16:00 to Monday 09:00 above zero	It covers only Bitcoin moves the stocks could not trade on in regular hours, ends in time to place an order, and has no threshold to tune.	Using the 09:30 price or an unfinished candle would be look-ahead bias. If only large moves mattered, a zero threshold would hide them, but the +1% and +2% checks did worse.

Choice	Why	If the assumption is wrong
Hold from open to close	It isolates the drift from the gap, and the exit is fixed in advance.	If the delayed reaction lasted longer than one day, a same-day exit would miss it.
Costs: 10 bps per side	An assumed all-in cost of spread and slippage for liquid stocks at the open, tested from 0 to 50 bps.	If real costs at the open are higher, returns fall quickly. At 50 bps the signal's out-of-sample return is −3.5%.
Benchmarks: every Monday and random Mondays	Buying every Monday has the same exposure, so a difference reflects timing. Random selections with the same number of trades separate timing from trading less.	Comparing only with cash would credit the signal with the basket's general rise in 2024.

Table 4: Main design choices

5. Pitfall Checklist

Look-ahead bias

Checked: The signal uses only Bitcoin candles that had finished by 09:00, half an hour before the trade at the 09:30 open (Section 2.2). Predictors and outcomes are kept in separate tables, and the signal function stops with an error if it receives any outcome column. As a direct test, the signal was rebuilt for all 1,195 eligible events with every candle after each event's cutoff hidden, and the returns and trade decisions were identical (notebook, Section 5). Monday prices and the Monday Bitcoin return are used only as outcomes or labelled hindsight checks.

Found: No look-ahead in the signal or the trade timing.

Survivorship bias

Checked: The universe is a fixed list of three companies chosen in 2026, when it was already known that all three came through the 2022 crypto crash and became the best-known Bitcoin-linked stocks. Miners that went bankrupt in that crash are not included. One example is Core Scientific, then the largest Bitcoin miner by computing power, which filed for Chapter 11 bankruptcy on December 21, 2022. It emerged and relisted in January 2024, but shareholders who held through the bankruptcy suffered losses that a survivors-only list leaves out.

Found: The bias exists. Including failed companies would probably have lowered the basket's returns, especially in 2022, so the absolute returns here are likely too high. The effect on the main question is smaller, since it concerns how quickly the open absorbs Bitcoin moves rather than which companies survived. The results apply only to these three stocks.

Overfitting/data-snooping

Checked: Only one trading rule was set before any results were seen. It buys when the Bitcoin return is above zero, so there is no number to tune. In total, three versions of the strategy were run (the main rule and two alternative thresholds of +1% and +2%, both chosen in advance as checks), together with the every-Monday benchmark and five cost levels. No version was added or dropped because of its results.

Found: In-sample, the +1% and +2% thresholds looked better than the primary rule (+3.2% and +14.2% against −18.8%), but the rule was not changed. Out-of-sample they did worse (+14.0% and +15.0% against +47.5%), so switching would have picked the weaker rule. Analyses added after the

test (the random-selection benchmark and the independence check) are labelled post-freeze and did not change the rule.

Transaction costs and slippage

Checked: Every buy and every sell costs 10 bps, so the trader buys slightly above the daily Open and sells slightly below the daily Close. Days in cash cost nothing. The same backtest was rerun with costs from 0 to 50 bps per side.

Net cumulative return	0 bps	5 bps	10 bps	25 bps	50 bps
Out-of-sample, BTC signal	64.0%	55.5%	47.5%	25.8%	−3.5%
Out-of-sample, every Monday	72.1%	57.1%	43.5%	9.2%	−30.7%
In-sample, BTC signal	−8.5%	−13.8%	−18.8%	−32.2%	−49.8%
In-sample, every Monday	3.7%	−8.8%	−19.7%	−45.3%	−71.2%

Table 5: Net cumulative return by cost per side.

Found: Costs decide the comparison. Before costs the benchmark is ahead in both periods; from about 10 bps the signal is ahead only because it trades about half as often. The 10 bps figure is an assumption, not a measured spread, and the daily Open and Close may not equal prices actually achievable in the opening and closing auctions. Market impact and capacity are not modelled, and fractional shares are allowed.

Out-of-sample validation

Checked: In-sample was 1 January 2021 to 31 December 2023 and out-of-sample was 1 January 2024 to 31 December 2025. The design was frozen in a git commit (fec2b54), and the out-of-sample period was evaluated once after that. One correction was logged: the outcome table was calculated for all years during data cleaning, so 2024–2025 outcomes were stored early, but none were displayed, analysed or used before the freeze.

Found: The headline return changed a lot between periods (−18.8% to +47.5%), but the benchmark changed by a similar amount (−19.7% to +43.5%), so the change reflects the market rather than the signal. The main findings were the same in both periods. The gap slope was close to 1 (0.99 and 1.04), the drift showed no relationship with Bitcoin, and the signal did no better than random Mondays. A lag-1 autocorrelation check found no dependence between consecutive events; this does not prove independence, so the bootstrap intervals should be read as approximate.

6. Efficiency Touchpoint

Every stage was timed on the saved data (notebook, Section 7). With $B = 89{,}094$ Bitcoin candles, the most expensive step in the first version was not part of the strategy but the look-ahead check in Section 5. For each event it rebuilt a hash map of all candles up to that event's cutoff, so that later candles could not be seen. Each rebuild costs $O(B)$, so checking K events cost $O(K \cdot B)$, which took 8.3 seconds for 150 events, longer than the rest of the pipeline combined.

The fix uses the fact that candles are already sorted by end time. Binary search (`bisect_right`) accepts an upper index `hi`, so the search can be limited to the candles that ended before the cutoff without rebuilding anything. Each event then needs two searches of $O(\log B)$, after a one-off $O(B)$ step to list the complete candles. The check became $O(K \log B)$ and ran in 0.0013 seconds on the same 150 events, about 6,000 times faster. Because it is now cheap, it covers all 1,195 eligible events instead of a sample. The signals it produces were confirmed to be identical.

Stage (after the fix)	Seconds	Complexity
Parse and clean Bitcoin files (includes sort)	0.27	$O(B \log B)$
Event alignment (hash map build and lookups)	0.23	$O(B + W)$
Correctness checks, all events	0.09	$O(K \log B)$ for the look-ahead check
Backtest, in-sample, both strategies	0.07	$O(D + T \cdot N)$
Bootstrap, 2,000 resamples	0.03	$O(R \cdot W)$
Look-ahead check, original version (150 events)	8.28	$O(K \cdot B)$

Table 6: Measured run times on the saved data. B candles, W events, K events checked, D trading days, T trades, $N = 3$ stocks, R resamples.

The most expensive step is now parsing and cleaning the Bitcoin files, at about 0.3 seconds. Reading the files is linear in B and the sort is $O(B \log B)$.

The same lookup problem shows a trade-off between the two data structures. For the 2,608 endpoint prices in the main pipeline, binary search took about 0.006 seconds and the hash map about 0.1–0.15 seconds, almost all of it spent building the dictionary; a naive scan through the candles would take about 15 seconds ($O(W \cdot B)$), estimated from 200 targets sampled across the whole period. A hash map is the better choice when many exact-match lookups are made on one fixed set. Binary search on sorted data is better when the data is already sorted or when the question is "the latest value at or before time T", which is why the look-ahead check needed it.

Downloading is the largest cost overall about 60 seconds for all 61 files. This is why the pipeline saves the data once and runs offline afterwards.

7. Risk Touchpoint

Sizing and leverage. The strategy never borrows. On a trade day it puts 100% of capital into the basket, one third in each stock, and on all other days it holds cash. Positions are held only from the open to the close, so there is no overnight or weekend risk, but each trade day carries the full basket's daily risk.

Concentration. Three stocks give little diversification, because all three are driven mainly by Bitcoin. During Monday the basket moved about 1.7 times Bitcoin's own move (Section 3), and MSTR, the most volatile name in 2024, dominates it.

Bad scenario. The main risk is a positive Bitcoin weekend followed by a reversal on Monday. The worst out-of-sample day was 3 March 2025. On Sunday 2 March, President Trump posted on Truth Social that a planned US "Crypto Strategic Reserve" would include XRP, SOL and ADA, and later added BTC and ETH. Bitcoin jumped from about \$84,000 to roughly \$94,000. The signal bought, but on Monday the rally reversed when Trump confirmed that 25% tariffs on Canada and Mexico would start the next day. Bitcoin fell about 8% to below \$85,000, the S&P 500 had its worst day of 2025, and the basket lost 12.3% by the close. A -10% basket day costs about 10.2% of capital once costs are included, and a few such days would erase most of a year's gains (removing the five best days already turns $+47.5\%$ into -14.3%). A Binance outage, a break in USDT's price or wide opening spreads could also make real results worse than the backtest, and large orders would move prices at the open, so the results only hold for small amounts of money.

8. Reflection

When I started, I thought of the idea simply as "weekend Bitcoin moves should show up in crypto stocks on Monday". The most useful thing I learned is that this sentence hides two different questions. Whether Bitcoin moves the opening price is almost obvious, and it turned out to be

true, since the gap followed Bitcoin almost one-for-one. So, whether anything is left after the open is the only question a trader can actually use, and the answer was no. I only properly understood why the gap cannot be traded, and why the signal has to stop at 09:00 rather than 09:30, once I worked through the $100, $106, $107 example step by step.

The data took more care than I expected. I assumed downloading prices would be the easy part, but the Binance files switched timestamp units in 2025, five candles had been cut short by exchange outages, and US and UK clocks change on different dates. Each of these could have quietly shifted a signal by an hour or used a wrong price if it had not been checked. The efficiency result also surprised me. The slowest part of my pipeline was a test, not the strategy, and limiting a binary search with one extra argument made it much faster.

My biggest wrong assumption was about what a good result looks like. The test period showed +47.5% against +43.5% for the benchmark, which at first looked like a success, and after seeing the results I even asked whether I should look for a different hypothesis that would be profitable. The random-selection test showed that the signal was no better than picking the same number of Mondays at random, and that its advantage came from trading less often. The +1% and +2% thresholds showed the same thing, because they looked better in-sample and worse out-of-sample, so changing the rule after seeing the data would have made it worse. I also had to correct my own description of the out-of-sample lock, and after that I checked each claim in the report against what the code does.

If I continued, I would first test the frozen rule unchanged on 2026 data, which played no part in this study. I would also widen the universe to include miners that failed or were delisted, use intraday stock prices to see how quickly the adjustment happens after 09:30, and replace the assumed 10 bps with measured opening-auction costs. Given how completely the open absorbed the weekend move, I do not expect these to make the strategy profitable, but they would show whether the conclusion holds beyond three well-known stocks.

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